Circuit model and solder practice

antocciphysics

UNIT 8 / DAY 2 / HONORS

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Test a circuit representation before making permanent connections.

SUCCESS CRITERIA - I can...

1. Build a complete battery-switch-motor circuit model.
2. Explain an open versus closed path.
3. Document a supervised solder-practice check.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A car receives 50 J electrical input but gains only 30 J translational kinetic energy. Name two possible destinations of the difference and explain why the account needs them.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $I = \Delta q/\Delta t$; $I = 0$ in an ideal open circuit.
Use the assigned simulation and teacher solder procedure. Disconnect power; hot irons/joints stay in the designated work area.

Worked reasoning: In a simple car circuit, battery positive connects through the switch and motor back to battery negative. A wire directly across the battery bypasses the motor and can create a hazardous short circuit.

Your turn: Draw the intended closed loop and circle a place where an open switch safely breaks it.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw your simulated closed circuit and an open-switch version. Predict which runs the motor and why.



Circuit symbols: long/short lines = battery; rectangles = resistors; M = motor.

Marzano check - circle: ① ② ③

04 Apply the physics

A test resistor has 6 V across it and carries 0.30 A. Calculate resistance and power. Compare it with the motor circuit shown: why might a running motor fail to behave like that fixed resistor?



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Make a two-column record: practice-joint observation and next improvement. Include teacher feedback on the joint and tool handling.

Observed joint / handling	Teacher feedback	Next time
___	___	___
___	___	___

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Two equal 12 Ω resistors are connected in parallel. Find equivalent resistance, then compare with a series connection of the same pair.
Draw the stated circuit and label quantities relevant to the question.

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

07 Exit check

Explain why a visually shiny joint alone is not enough evidence of a reliable electrical and mechanical connection.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 6 V motor circuit is modeled with 3 ohms while running and 1 ohm at stall. Compare currents and input powers; explain why resistance/current may change with motion and why the model is simplified.



Marzano check - circle: 1 2 3